

CADENCE DESIGN SYSTEMS, INC. (NASDAQ: CDNS)

# The Tollbooth on Every AI Chip, Priced Past Perfection

Equity Research Note · Technology — Application Software (EDA) · October 4, 2026

| Recommendation | 12-Mo. Price Target | Current Price                  | Implied Upside | Risk Rating                              |
|----------------|---------------------|--------------------------------|----------------|------------------------------------------|
| <b>REDUCE</b>  | <b>\$240</b>        | <b>\$351.35</b><br>Oct 2, 2026 | <b>−31.7%</b>  | <b>Medium</b><br>Duopoly, China overhang |

| Snapshot              |                        |                            |                               |
|-----------------------|------------------------|----------------------------|-------------------------------|
| Market cap            | ~\$96.8 bn             | Shares (diluted)           | ~275 mn                       |
| Enterprise value      | ~\$97.9 bn             | 52-week range              | \$262.75 – \$416.69           |
| Net debt              | ~\$1.1 bn              | Dividend                   | None                          |
| 2026E revenue (guide) | \$6.26 – \$6.34 bn     | 2026E non-GAAP EPS (guide) | \$8.05 – \$8.15               |
| Q2'26 revenue         | \$1,584 mn (+24.2%)    | Q2'26 non-GAAP op margin   | 45.5% (+270 bps)              |
| Next catalyst         | Q3'26 earnings, Oct 26 | Analyst consensus          | Moderate Buy, avg PT \$403.88 |

This note is independent research-style analysis for informational purposes only and is not investment advice. Figures are drawn from Cadence's SEC filings and earnings releases and reputable financial press as of October 2, 2026. Projections and the price target are the author's estimates under a hardened 10-year scenario-DCF methodology.

# 1 Investment Thesis

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Cadence is one of the finest software businesses on earth: a ~30% share of the electronic-design-automation duopoly, ~86% gross margins, ~80% recurring revenue, a record \$8.1B backlog, and four straight quarters of beat-and-raise execution as AI-chip design activity accelerates. Q2 2026 was outstanding — revenue +24.2% to \$1.584B, non-GAAP operating margin 45.5% (+270 bps), full-year guidance raised for the second time to ~\$6.3B (+19%). None of that is in dispute. The problem, as always with this name, is the price. At \$351 the stock trades at 35.9x forward non-GAAP earnings and 16.7x EV/revenue, and our reverse DCF shows the market is capitalizing roughly **17.5% annual free-cash-flow growth for a full decade** — a trajectory *above our bull case*. We have no room to pay for the two real overhangs: China/export controls and the decade-long question of what agentic AI does to EDA pricing power.

The hardened rebuild moves us from HOLD to **REDUCE**. Our 10-year scenario DCF — weights bear 25% / base 50% / bull 25%, discount rates 11.5% / 9.0% / 8.0%, terminal growth capped at 2.5% on normalized mid-cycle margins — yields a probability-weighted fair value of **\$240, 31.7% below** the current price. Even the bull case (\$382, after a disclosed terminal-value haircut) is just 8.6% above the quote; the bear case (\$63, –82%) is genuinely painful. The target is the weighted DCF: \$240. Holders should reduce — this is the second time this name has traded at better-than-bull perfection in twelve months, and perfection has a poor record of persisting.

## Audit of the October 2 note (red-team lens)

The prior note said HOLD with a \$300 target, dated October 2, 2026. Its analysis of the business was sound; its valuation leaked optimism through **4–5.5% terminal growth** in the scenario DCF — a perpetuity assumption that compounds into most of the enterprise value and that no mature software franchise sustains. Its bear case (\$137) was mild enough to be described, not felt, and its target was set *between* the weighted DCF (\$276) and through-cycle multiples — a judgment departure the hardened methodology forbids. This rebuild corrects all three: terminal growth ≤2.5%, a bear that hurts (\$63, –82%), and a target that equals the weighted DCF unless a written, falsifiable justification departs from it. There is no such justification here. The three strongest arguments against owning the stock at \$351: **(1)** the quote prices 17.5% annual FCF growth for a decade, above our 16.5% bull case — there is no upside left to pay for; **(2)** China/export controls are not a tail risk but a priced-in certainty — a July 2025 guilty plea, \$140M in penalties, federal probation to ~2028, and 12–15% of sales exposed to further BIS tightening; **(3)** Cadence trades at 35.9x forward earnings versus Synopsys at 26.4x, despite near-identical scale — the market pays Cadence as though the duopoly has one winner while the Ansys-armed Synopsys gets stronger. The load-bearing assumption of every bullish case is that mid-teens revenue growth persists for most of a decade *at peak margins*; the hardened rebuild tests the price on normalized margins and 2.5% terminal growth, and the price fails the test.

## Why it works (bull)

**Duopoly tollbooth.** ~30% of EDA; you cannot tape out a modern chip without Cadence or Synopsys.

**Switching costs are extreme** — flows qualified over years across foundry PDKs, tens of thousands of tool-specific scripts, and a failed 2nm tapeout costs tens of millions.

**AI supercycle.** 2nm/A16 ramp, chiplets/3DFabric, hyperscaler in-house silicon — all EDA-intensive. IP grew 40%+ in Q2 (Intel agreement); System Design & Analysis +37%; hardware posted a record quarter.

**Rare visibility.** \$8.1B backlog (\$4.2B in the next 12 months) covers ~2/3 of the \$6.3B guide — three-year all-you-can-eat token deals make ~80% of revenue recurring.

**Fortress economics.** 45.5% non-GAAP operating margin, ~\$2B of 2026 operating cash flow, ~50% of FCF directed to buybacks, net debt of just ~\$1.1B (~0.5x EBITDA).

## What breaks it (bear)

**Priced past perfection.** 17.5% perpetual FCF growth is in the price — above our bull case. A single guide-down quarter would likely cost 15–20% before fundamentals changed; the stock already de-rated 18% from its \$416.69 high and is still priced for flawlessness.

**China/export controls.** July 2025 guilty plea on export-control conspiracy (\$45.3M of technology sold to entities tied to Chinese military supercomputing), \$140M in penalties, three years of federal probation to ~2028 with mandated compliance reporting. China is ~12–15% of sales; tightened BIS EDA restrictions directly cap the growth algorithm in one of the world's largest chip-design markets.

**Agentic AI and the seat-license question.** Near-term, AI agents expand EDA compute consumption (bullish). Long-term, if AI compresses design cycles and engineer headcount, the per-seat licensing model faces its first existential question in 30 years. The Kimi K3 episode (July 2026) showed how fast the market prices this narrative.

**Competition.** Synopsys closed the \$35B Ansys acquisition in July 2025 — the strongest rival Cadence has ever faced (EDA + multiphysics simulation under one roof), currently digesting a mega-merger. Cadence at 16.7x EV/revenue vs SNPS at 10.8x prices Cadence as the undisputed winner of the AI cycle.

## 2 Business Overview — What Cadence Does

Cadence Design Systems (San Jose, CA; founded 1988) sells the computational software, emulation hardware, and semiconductor IP that chipmakers and systems companies use to design, verify, and tape out chips. Roughly 60% of all EDA spending flows through Cadence (~30%) and Synopsys (~31%), with Siemens EDA (~13%) a distant third. The business runs in four product groups, all of which grew double-digits in Q2 2026: **Core EDA** (+18%) — the digital full flow (Genus, Innovus, Tempus, Quantus, Voltus) and the Virtuoso custom/analog platform, the industry standard, with Cerebrus AI-driven optimization; **Functional verification & hardware** (record quarter) — Palladium emulation, Protium FPGA prototyping, Xcelium simulation, Jasper formal verification; **Semiconductor IP** (+40%+) — interface IP (PCIe, SerDes, DDR/LPDDR, UCIe for chiplets), memory compilers, Tensilica/DSP cores, the highest-growth, highest-leverage line as chiplet architectures proliferate; **System Design & Analysis** (+37%) — Allegro/OrCAD PCB plus multiphysics (Clarity, Celsius, Sigrity, Fidelity CFD, BETA CAE) selling 'beyond silicon' to hyperscalers, auto, and aerospace.

The 2026 strategic overlay is AI on both sides of the transaction: 'design for AI' (TSMC N3/N2/A16/A14 certification, 3DFabric chiplet flows for AI accelerators) and 'AI for design' (the agentic AI Super-Agent portfolio, which management says is seeing strong early traction and should drive higher EDA compute consumption). Whether agentic AI is ultimately a volume tailwind or a pricing headwind is the central long-term debate — and it is precisely why we refuse to pay a terminal growth rate above 2.5%.

### Financial situation

A compounding machine with software economics: gross margins have held 86–90% for a decade, revenue has compounded ~13% annually, and the model is ~80% recurring. GAAP operating margins sit near 30%; non-GAAP, which strips out stock comp and acquisition amortization, runs 43–46% — the number management guides and the Street models. Note the 2022–24 net-income plateau (\$0.85B → \$1.06B) against 30% revenue growth — BETA CAE acquisition amortization, restructuring, and SBC absorbed the operating leverage on a GAAP basis.

| Fiscal year          | 2022   | 2023   | 2024   | 2025   | 2026E (guide) |
|----------------------|--------|--------|--------|--------|---------------|
| Revenue (\$B)        | 3.56   | 4.09   | 4.64   | 5.30   | 6.26–6.34     |
| Revenue growth       | +18.8% | +14.8% | +13.5% | +14.1% | ~+19%         |
| Non-GAAP op margin   | ~42%   | ~43%   | ~43%   | ~44%   | ~44–45%       |
| Free cash flow (\$B) | 1.12   | 1.25   | 1.12   | 1.59   | ~2.0          |

Q2 2026 (reported July 27) was a clean beat-and-raise — the fourth consecutive quarter of that pattern. Revenue of \$1.584B (+24.2% YoY) landed at the top of the \$1.555–1.595B guide; non-GAAP EPS of \$2.11 (+27.9%) beat the \$2.05–2.06 consensus; non-GAAP operating margin hit 45.5% (+270 bps YoY); billings were \$1.73B (+31.5%). Management raised the full-year outlook for the second time: revenue \$6.26–6.34B (from \$6.125–6.225B) and non-GAAP EPS \$8.05–8.15 (from \$7.85–7.95). The balance sheet is a fortress: \$1.44B cash against \$2.5B debt (~\$1.1B net debt, ~0.5x EBITDA), with ~50% of ~\$2B 2026 FCF directed to buybacks.

## 3 Valuation — Hardened 10-Year DCF, Fair Value \$240

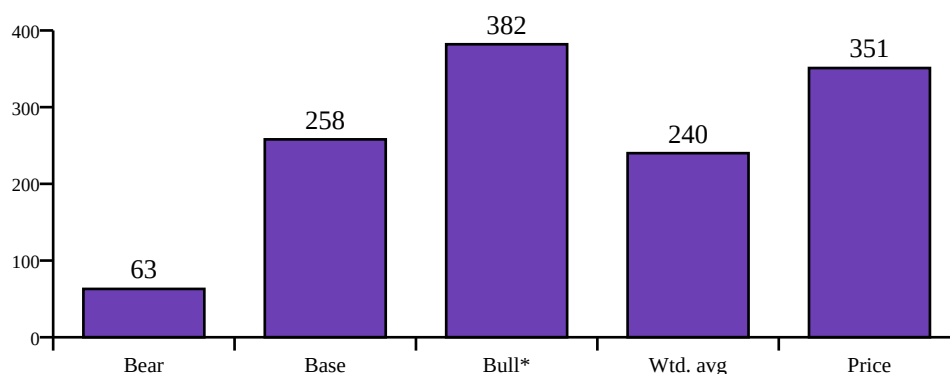
The prior note's 5-year DCF ran terminal growth of 4–5.5% — a perpetuity assumption that did the load-bearing work. The hardened rebuild values Cadence on a **10-year scenario free-cash-flow DCF** — weights bear 25% / base 50% / bull 25% — with scenario-specific discounts: **bear 11.5%** (base + 250bp), **base 9.0%** (stable-compounder tier: duopoly tollbooth, ~80% recurring revenue, record \$8.1B backlog), **bull 8.0%** (base – 150bp, at the floor). Terminal growth is **2.5% / 2.5% / 1.0%**, applied to year-10 FCF at **normalized mid-cycle margins (33% / 36% / 28%)** — deliberately below the 36.8% Q2 print, i.e. never peak margins. Growth inputs are independently justified: the base case's 13% revenue CAGR is Cadence's own decade-long historical rate (no management guidance taken at face value); the bear's 2% and the bull's 16.5% bracket the China-shock and AI-supercycle outcomes.

### Scenario DCF (10-year, from 2026E revenue \$6.30B):

| Scenario                                              | 2027–36 rev CAGR | 2036 revenue | FCF margin (yr-10) | Discount / term. g | Fair value       |
|-------------------------------------------------------|------------------|--------------|--------------------|--------------------|------------------|
| Bear: China shock, EDA digestion                      | 2.0%             | \$7.7 bn     | 28%                | 11.5% / 1.0%       | <b>\$63.35</b>   |
| Base: AI tailwind, mid-teens compounding              | 13.0%            | \$21.4 bn    | 33%                | 9.0% / 2.5%        | <b>\$257.76</b>  |
| Bull: AI supercycle, share gains                      | 16.5%            | \$29.0 bn    | 36%                | 8.0% / 2.5%        | <b>\$381.69*</b> |
| <b>Probability-weighted (25/50/25) → \$240 target</b> |                  |              |                    |                    | <b>\$240</b>     |

\*Bull-case fair value is **after a disclosed 20% terminal-value haircut**: the raw bull DCF (\$447) had the terminal value at 73% of enterprise value, above our 70% guardrail. Even after the haircut the bull offers only +8.6% above the quote — the entire distribution lives below the price. The bear hurts on all three required axes: revenue CAGR 2.0% (<3%), FCF margins 28% vs 33% base (500bp compression), and the 250bp derating (11.5% vs 9.0%). Bear fair value \$63.35 sits 82% below the price.

### Scenario fair values vs. current price (\$)



\*Bull shown post-haircut.

### Reverse DCF: what is priced in?

At \$351.35, enterprise value is ~\$97.9B. Holding a 9% discount rate and 2.5% terminal growth, the price implies roughly **17.5% annual free-cash-flow growth for a full decade** — a trajectory *above our 16.5% bull-case revenue CAGR*. The prior note's 5.9% perpetual-growth framing understated the embedded optimism by measuring it the wrong way: the market is not capitalizing a modest perpetuity, it is underwriting a decade of superlative growth *and then* a 2.5% perpetuity. There is no margin of safety whatsoever; any combination of China deceleration, EDA digestion, or multiple compression breaks the math.

**Price target: \$240.** The target *is* the probability-weighted hardened DCF —  $0.25 \times \$63.35 + 0.50 \times \$257.76 + 0.25 \times \$381.69$  — with no multiple overrule and no consensus anchoring. Implied downside -31.7%. What would falsify it: China revenue growing through the export-control overhang (proving the 12–15% exposure is ring-fenced and the probation is a non-event), or 2027 revenue growth printing above 20% again — either would argue the bull case deserves to be the base case.

## 4 Risks & Catalysts

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### Risks

**China and export controls (the real overhang).** Guilty plea July 2025, \$140M penalties, federal probation to ~2028 with mandated compliance reporting. China has historically been ~12–15% of revenue; tightened BIS EDA restrictions directly cap the growth algorithm.

**Valuation / multiple compression.** 35.9x forward non-GAAP EPS, 16.7x EV/revenue, PEG 2.5 — the stock de-rated 18% from its \$416.69 high and still prices 17.5% annual FCF growth for a decade.

**Agentic AI and the seat-license question.** The Kimi K3 episode (July 2026) was a preview of how fast the market prices the narrative that AI compresses design cycles and seats.

**Competition.** The Ansys-armed Synopsys is the strongest rival Cadence has ever faced; share shifts in EDA are measured in single points per year — but they compound.

**Customer concentration & hardware lumpiness.** Hyperscalers and top-tier semis dominate bookings; Palladium hardware quarters can swing reported growth by points.

### Catalysts

**Q3 2026 earnings (Oct 26):** the next beat-and-raise would extend the streak to five quarters; watch backlog conversion and China commentary.

**TSMC N2/A16 ramp and 3DFabric adoption:** each leading-edge ramp is a multi-year EDA/IP annuity; the Intel IP win suggests share gains at the frontier.

**Buybacks:** ~50% of ~\$2B 2026 FCF directed to repurchases — a put under the stock on weakness.

**SNPS-Ansys integration friction:** mega-merger digestions historically open share windows; Cadence's clean execution is its best weapon.

## 5 Recommendation

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**REDUCE, \$240 target (-31.7%).** Cadence is the highest-quality name in EDA — a duopoly tollbooth with 86% gross margins, 45.5% non-GAAP operating margins, a record \$8.1B backlog, and the AI-chip cycle's purest picks-and-shovels exposure outside the foundries. We would love to own it. We refuse to pay \$351 for it when the quote capitalizes 17.5% annual FCF growth for a decade — better than our bull case — and the Street's \$404 target demands the bull case as the base case. The hardened 10-year DCF values the equity at \$240 (bear \$63.35 / base \$257.77 / bull \$381.69 post-haircut) against a \$351.35 quote; even the bull case is only 8.6% above the price. The prior note's \$300 target rested on 4–5.5% terminal growth and a mild bear case; the rebuild rejects both.

This is a price call, not a business call. The business is a compounder; the stock is a fully-priced one, now past perfection. **Current holders should reduce** — the position is a momentum trade wearing a compounding story's clothes, and the China/probation overhang has not gone away. We would upgrade to HOLD toward our **\$240** fair value and to BUY below ~\$200, where even the bear case's pain is more than compensated. **Falsification:** China revenue growing through the export-control overhang, or 2027 revenue growth printing above 20% again — either would argue the bull case is the base case and force us to rebuild.

Disclosure: This note is independent research analysis for informational purposes only and is not personalized investment advice, a recommendation to transact, or an offer to buy or sell any security. Valuation models are estimates with wide confidence bands; multiples can remain elevated for years at quality compounders. Company data: Q2 2026 and FY2025 press releases, 10-Ks. Market data: Yahoo Finance (Oct 2, 2026). Consensus: MarketBeat/MarketScreener, Sep–Oct 2026 (17 analysts, avg target \$403.88). Export-control history: DOJ plea reporting, July 2025. The author holds no position in CDNS.